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Technical Presentation

DSA0304 – Natural Processing Language

Porter Stemmer: Design, Working Mechanism, and Applications

Presented By

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Introduction

- Stemming is a Natural Language Processing technique used to reduce words to their common root or stem.
- Porter Stemmer is a popular rule-based stemming algorithm developed by Martin Porter in 1980.
- It removes common suffixes from English words using a sequence of predefined rules.
- For example, playing, played, and plays → play.
- It is widely used in information retrieval, search engines, text mining, and document classification.

Design of Porter Stemmer

Algorithm

- **Step 1:** Removes plurals and common endings.
- **Step 2:** Converts longer suffixes into shorter ones.
- **Step 3:** Removes additional suffixes.
- **Step 4:** Removes larger suffixes.
- **Step 5:** Performs final cleanup.

Input Text
↓
Tokenization
↓
Porter Stemmer
↓
Suffix Removal
↓
Stemmed Words

Working Mechanism

Input Text
↓
Tokenization
↓
Identify Suffix
↓
Apply Stemming
↓
Remove/Modify Suffix
↓
Generate Stem
↓
Output

Example

playing
↓
Identify suffix: "ing"
↓
Apply stemming rule
↓
Remove "ing"
↓
play



Applications



- Search engines : Matches different forms of words to improve search results.
- Information retrieval :Helps retrieve relevant documents based on related word forms.
- Text classification :Helps categorize text by reducing word variations.
- Document clustering : Groups similar documents using common word stems.
- Text mining : Helps analyze word frequency and discover patterns in text.
- NLP preprocessing : Simplifies text before further language processing.





Advantages

- Simple and fast.
- Easy to implement.
- Reduces vocabulary size.
- Improves word matching in search systems.

Limitations

- May produce non-English stems.
- Can cause over-stemming.
- Can cause under-stemming.
- Mainly designed for English.

Conclusion

Porter Stemmer is an efficient rule-based stemming technique that reduces different forms of words to a common stem, making it useful for NLP and information retrieval systems.


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